

TIC 367157113

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 0.728977 d (ALIASED, see below)
-> corrected period: 1.457953 d
Epoch (t0) 2500.7803 BTJD
Duration 0.3500 d
Depth 2.543 % (25,427 ppm)
SNR 1940.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

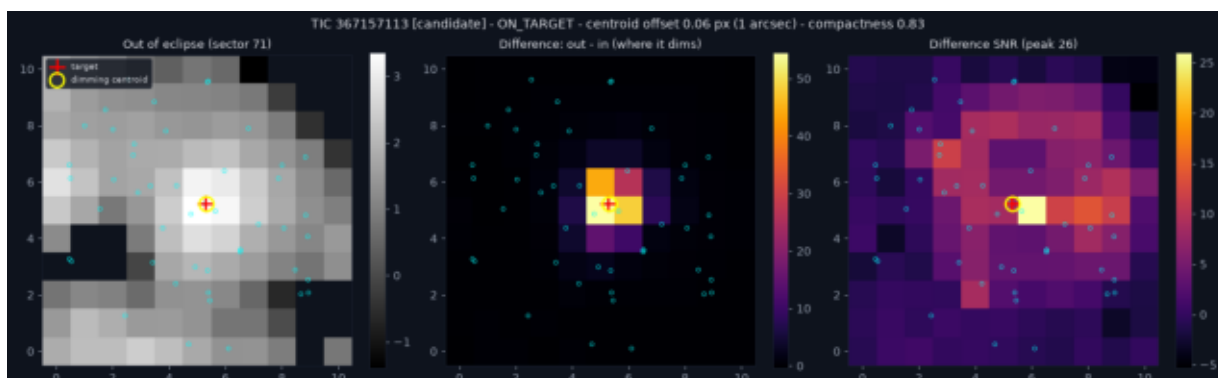
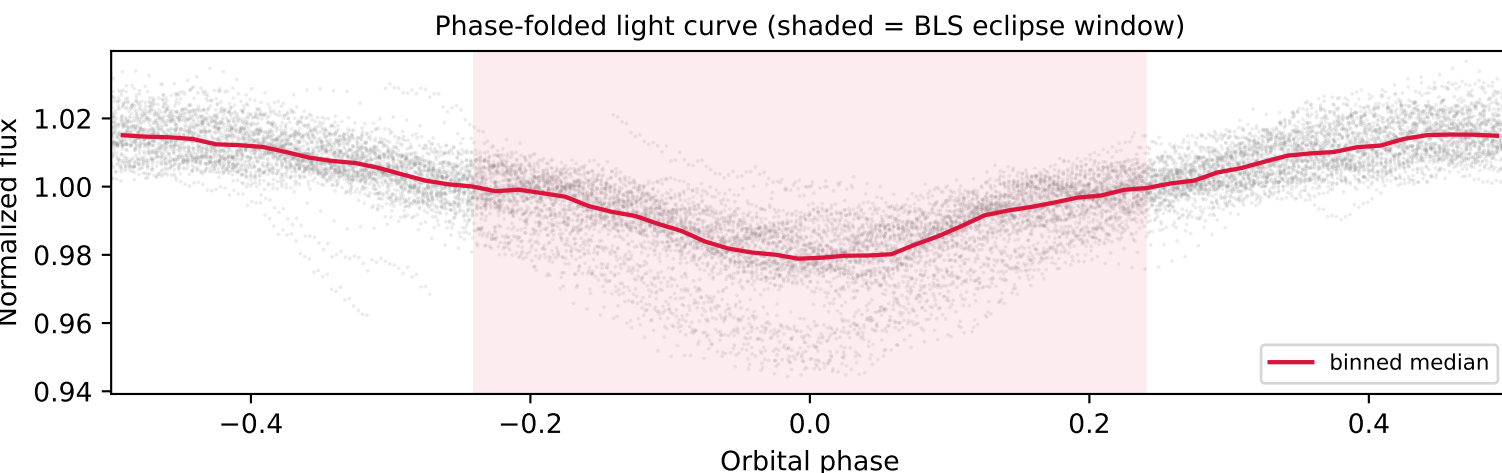
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 93 (gate ≥ 3)
Robust depth ratio 0.78 (gate ≥ 0.50)
Robust depth sigma 163.6 (gate ≥ 7.0)
Odd/even z-score 19.25 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 71 light-curve dip in this sector: 98.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 26.03 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 19.25 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.457953 d, in any write-up.